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National Institute of Technology Goa

Programme Name: **B.Tech.**

Minor Examination-I, 2025-26

Course Name: **Differential Equations and Vector Calculus**

Date: **19/02/2025**

Duration : **1 Hour**

Course Code: **MA150**

Time: **11:00 AM to 12:00 NOON**

Max. Marks: **25**

1. Answer All Questions.

2. No marks will be given if the explanation of your answer is missing.

1. Solve the differential equation $x^2 dx + (3y^2 - x^3 y^2) dy = 0$. [5 Marks]

2. Solve the differential equation $(2x^2 y^2 + y) dx - (x^3 y - 3x) dy = 0$. [5 Marks]

3. Find the volume inside the capsule bounded by the paraboloids [5 Marks]

$$z = 9 - x^2 - y^2 \quad \text{and} \quad z = 3x^2 + 3y^2 - 16.$$

4. Evaluate [5 Marks]

$$\iint_D (x^2 - y^2) e^{xy} dx dy,$$

where D is the region in the first quadrant bounded by the hyperbolas

$$xy = 1, \quad xy = 4,$$

and the lines

$$y = x, \quad y = x + 2.$$

Use the change of variables:

$$u = xy \quad \text{and} \quad v = y - x.$$

5. Consider the cone of radius a and height h and, in this case the cone is the solid W bounded by the surface

$$az = h\sqrt{x^2 + y^2}$$

and the plane

$$z = h.$$

Use cylindrical coordinates to calculate its volume and also compute the centroid of W . [5 Marks]

Roll No 25CSE1049



National Institute of Technology Goa

Programme Name: B.Tech.

Minor Examination-II, 2025-26

Course Name: Differential Equations and Vector Calculus

Date: 18/04/2026

Duration : 2 Hours

Course Code: MA150

Time: 10:00 AM to 12:00 NOON

Max. Marks: 50

1. Answer All Questions.
2. No marks will be given if the explanation of your answer is missing.

1. Use Stokes' theorem to evaluate $\int_C F \cdot dr$ if $F = xzi + xyj + 3xzk$ and C is the boundary of the portion of the plane $2x + y + z = 2$ in the first octant, traversed counterclockwise as viewed from above. [10 Marks]

2. State the divergence theorem and use to evaluate

$$\iint_{\partial W} (x^2 + y + z) dS,$$

where ∂W is the surface of the sphere $x^2 + y^2 + z^2 \leq 1$.

[8 Marks]

3. Evaluate $\iint_S \mathbf{F} \cdot d\mathbf{S}$, where

$$\mathbf{F}(x, y, z) = xy^2 \mathbf{i} + x^2 y \mathbf{j} + y \mathbf{k}$$

and S is the surface of the cylinder $x^2 + y^2 = 1$, bounded by the planes $z = 1$ and $z = -1$, and including the portions $x^2 + y^2 \leq 1$ when $z = \pm 1$. [8 Marks]

4. Solve the differential equation $(x^2 - 2x + 2y^2) dx + 2xy dy = 0$. [8 Marks]

5. Find the integral of the equation $\frac{d^2x}{dt^2} + 2n \cos \alpha \frac{dx}{dt} + n^2 x = a \cos nt$, which is such that when $t = 0$, $x = 0$ and $dx/dt = 0$. [8 Marks]

6. Solve the differential equation $y'' - 4y' + 8y = e^{2t}$, $y(0) = 2$, $y'(0) = -2$ using Laplace transform. [10 Marks]

ALL THE BEST

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National Institute of Technology Goa

Programme Name: B.Tech.

Mid Semester Examinations, March 2026

Course Name: **Differential Equations and Vector Calculus**

Course Code: **MA150**

Date: 06/03/2026

Time: 04.00 PM - 05.30 PM

Duration: 1 Hour 30 Minutes

Max. Marks: 50

ANSWER ALL QUESTIONS

Part A (Ordinary Differential Equations)

1. Solve the differential equation $(2x^2y - y - ax^3) dx + (1 - x^2) x dy = 0$, $a \in \mathbb{R}$. [5]
2. Find the all values of a and b for which the differential equation $y'' + 2ay' + by = 0$, $a, b \in \mathbb{R}$ has periodic solutions. [5]
3. Let $y = f(x)$ is the solution of $p(x)y'' + q(x)y' + r(x)y = 0$, and $y = C_1f(x) + C_2v(x)f(x)$ is the general solution of given equation then find $v(x)$. [5]
4. Find the complete solution of the differential equation $(D^2 + 2)y = x^2e^{3x} + e^x \cos 2x$. [5]
5. Let $y = e^{ax}$ is the solution of $xy'' - (2x - 1)y' + (x - 1)y = 0$. Then find the value of a and also the second linearly independent solution. [5]

Part B (Multiple Integrals)

6. Evaluate $\iint_D e^{x/y} dA$, where D is the region between the curves $y = x$ and $x = y^3$ in the first quadrant.
7. Evaluate the integral with suitable substitutions

$$\int \int_R e^{\frac{x-y}{x+y}} dA,$$

where

$$R = \{(x, y) : x \geq 0, y \geq 0, x + y \leq 1\}.$$

8. Find the volume of the solid region (ice-cream cone with a scoop of ice-cream) that lies inside the sphere $x^2 + y^2 + z^2 = z$ and above the cone $z^2 = x^2 + y^2$, $z \geq 0$.
9. Find the center of mass of the solid region W of constant density δ that is inside the sphere $x^2 + y^2 + z^2 = 1$ and above the cone $z^2 = x^2 + y^2$, $z \geq 0$.

10. Calculate the work done by the force

$$\mathbf{F} = x\mathbf{i} - y\mathbf{j} + (x + y + z)\mathbf{k}$$

on a particle that moves along the parabola $y = 3x^2$, $z = 0$ from the origin to the point $(2, 12, 0)$.

ALL THE BEST

Roll No 2 5 C 5 E 1 0 4 9



National Institute of Technology Goa

Programme Name: B.Tech
End Semester Examinations, 2025-26

Course Name: Differential Equations and Vector Calculus
Date: 08/05/2026
Duration: 3 Hours

Course Code: MA150
Time: 10:00 A.M - 01:00 P.M
Max. Marks: 100

1. Answer All Questions.
2. No marks will be given if the explanation of your answer is missing.
3. The question paper consists of **two** pages.
4. Write the answers to all the questions of each part in one place.

Part A

1. Consider the differential equation $(2xy^2 - 2y) dx + (3x^2y - 4x) dy = 0$.
(a) Check whether the given ODE is exact or not?
(b) If yes find its general solution.
(c) If not find the integrating factor and its general solution. [5M]
2. Reduce $2x^2y \frac{d^2y}{dx^2} + 4y^2 = x^2 \left(\frac{dy}{dx}\right)^2 + 2xy \frac{dy}{dx}$ homogeneous form by making the substitution $y = z^2$ and hence solve it. [5M]
3. Using the method of undetermined coefficients find the solution of $(D^2 + 1)y = 2 \cos x$. [5M]
4. Solve by using the method of variation of parameters $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} = e^x \sin x$. [5M]
5. For the initial value problem $\frac{dy}{dx} = e^y, y(0) = 0$. Find the largest interval $|x| \leq a$ in which the Picard's theorem guarantees existence of a unique solution. [5M]
6. Find the Laplace transform of $\frac{\cos \sqrt{t}}{\sqrt{t}}$. [5M]
7. Find the Laplace transform of periodic function $f(x)$ with period a . [5M]
8. Find inverse Laplace transform of $\log \left(\frac{s(s^2 + a^2)}{(s^2 + b^2)} \right)$. [5M]

9. Find inverse Laplace transform of $\frac{1}{s^3 + 3s^2 + 2s}$ using convolution. [5M]

10. Solve the integro-differential equation $\frac{dy}{dt} + 2y + \int_0^t y dt = 2 \cos t$, $y(0) = 1$ using Laplace transform. [5M]

Part B

11. Find the area of the playing field described by $R : -2 \leq x \leq 2$, $-1 - \sqrt{4 - x^2} \leq y \leq 1 + \sqrt{4 - x^2}$ using (a) Integration, and (b) Simple geometry. [5M]

12. Evaluate the integral $\int_0^1 \int_0^{\sqrt{1-x^2}} (x^2 + y^2) dy dx$. [5M]

13. Find the volume of the region D enclosed by the surfaces $z = x^2 + 3y^2$ and $z = 8 - x^2 - y^2$. [5M]

14. Find the centroid ($\delta = 1$) of the solid enclosed by the cylinder $x^2 + y^2 = 4$, bounded above by the paraboloid $z = x^2 + y^2$, bounded below by the xy -plane. [5M]

15. Show that $\mathbf{F} = (e^x \cos y + yz) \mathbf{i} + (xz - e^x \sin y) \mathbf{j} + (xy + z) \mathbf{k}$ is conservative over its natural domain and find a potential function for it. [5M]

16. Calculate the outward flux of the vector field $\mathbf{F}(x, y) = 2e^{xy} \mathbf{i} + y^3 \mathbf{j}$ across the square bounded by the lines $x = \pm 1$ and $y = \pm 1$. [5M]

17. Find the circulation of the field $\mathbf{F} = (x^2 - y) \mathbf{i} + 4z \mathbf{j} + x^2 \mathbf{k}$ around the curve C in which the plane $z = 2$ meets the cone $z = \sqrt{x^2 + y^2}$, counterclockwise as viewed from above. [5M]

18. Verify divergence theorem for the expanding field $\mathbf{F} = x \mathbf{i} + y \mathbf{j} + z \mathbf{k}$ over the sphere $z = x^2 + y^2 + z^2$. [5M]

19. Use Stokes' theorem to evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$, if $\mathbf{F} = xz \mathbf{i} + xy \mathbf{j} + 3xz \mathbf{k}$ and C is the boundary of the portion of the plane $2x + y + z = 2$ in the first octant traversed counterclockwise as viewed from above. [5M]

20. Solve

$$u_t = a^2 u_{xx}, 0 < x < L, t > 0,$$

$$u(0, t) = 0, u(L, t) = 0, t > 0,$$

$$u(x, 0) = f(x), -0 \leq x \leq L,$$

with $f(x) = x(x^2 - 3Lx + 2L^2)$.

[5M]

SMILE